Jane Yang

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Ph.D. student studying how people perceive and interact with objects in the real world, and using that to evaluate and inform vision models. I combine behavioral experiment design with large-scale work on child and adult egocentric video: detection and embedding pipelines, human-model benchmarks for object and 3D perception, and analyses of how naturalistic visual input differs from curated training data, now extending to object interactions and controlled training experiments. Interested in embodied perception and action, model evaluation, and how training data structure shapes what models learn.

Education

Ph.D. in Experimental Psychology, UC San Diego. Advisor: Bria Long — 2024–present

B.S. in Cognitive Science, UC San Diego. Machine Learning & Neural Computation; CS minor — 2018–2022

Technical Skills

Programming	Python, R, C++, JavaScript/TypeScript
ML & Vision	PyTorch, HuggingFace; CLIP, DINOv3, YOLOE/YOLOv8, Qwen VLMs, MediaPipe, OpenPose, WhisperX
Research Methods	Behavioral experiment design (web and in-lab), human-model benchmarking, representational similarity analysis, crowdsourced annotation, gold-set evaluation
Data & Infra	Large-scale video processing (FFMPEG), SLURM GPU clusters, Docker, Git, automated testing
Sensing & Hardware	Head-mounted eye tracking (Pupil Labs Neon), depth cameras (Orbbec Femto Bolt), IMU, multi-sensor sync

Experience

Graduate Researcher, UC San Diego Visual Learning Lab — 2024–present

- Built and maintain an end-to-end detection and embedding pipeline over 1,482 hours of egocentric video from 35 families: YOLOE detection, CLIP-based false-alarm filtering, crowdsourced validation, and CLIP/DINOv3 feature extraction on a GPU cluster; now moving detection to a local Qwen VLM, validated against a human-annotated gold set
- Quantified how object statistics in children's everyday visual input (frequency, exemplar variability, category geometry) diverge from adult egocentric experience (VEDB) and curated benchmarks such as THINGS, using representational similarity analysis over 85 categories; first-author manuscript under review, with a public reproducibility capsule (pinned Docker environment)
- Built and deployed MOCHI Kids, a developmental adaptation of the MOCHI 3D shape-oddity benchmark for human-model comparison on multiview object consistency (TypeScript/jsPsych, Express/MongoDB, automated end-to-end tests)
- Characterized within-category visual variability across 7,018 validated crops, using global and local dispersion metrics to separate viewpoint-driven from instance-driven variability; rankings largely agree across CLIP and DINOv3 and are only weakly tied to object frequency. Released as a reproducible package and interactive explorer (CCN 2026 spotlight)
- Designing controlled training experiments that manipulate viewpoint and instance variability to test how each shapes category learning and generalization in vision models
- Prototyping interaction measurement from detections, hand tracking, depth, and IMU head direction
- Evaluated CLIP image-text alignment on infants' visual and linguistic experience against curated caption datasets
- Designed a synchronized capture rig (two depth cameras, a security camera, two eye trackers) for caregiver-child play

Lab Technician, UT Austin Developing Intelligence Lab — 2022–2024

- Built Python and MATLAB pipelines to process, analyze, and visualize multimodal streams from eye trackers, cameras, IMUs, and microphones recorded during naturalistic play in homes
- Fine-tuned YOLOv8 for egocentric object detection, deployed MediaPipe and OpenPose for hand and face tracking, and integrated WhisperX for transcription and diarization

Selected Publications

- **Yang, J.**, Sepuri, T., Tan, A.W.M., Aw, K.L., Frank, M.C., Long, B. (2026). Characterizing the visual representation of objects from the child's view. *Under review.*
- Tan, A.W.M.*, **Yang, J.***, et al. (2025). Assessing the alignment between infants' visual and linguistic experience using multimodal language models. *Preprint*, arXiv:2511.18824.

Also *Developmental Science* (2025); co-organizer, *Beyond Curated Datasets* satellite, CCN 2026; invited talk, UNR Early Career Speaker Series (2026). Full list on Scholar.

Awards

NEI Early Career Scientist Travel Grant, Vision Science Society (2026); Anderson Travel & Research Award (2025) and Norman Henry Anderson Fellowship (2024), UC San Diego.